

Design and Implementation 8 bit CPU Architecture on Logisim For Undergraduate Learning Support

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Abstract— Computer Architecture consists of some basic design based on Computer Organization and Architecture (COA) such as logic design. Leading students to improve knowledge about COA needs a comprehensive learning with a working simulation of a simple 8-bit Central Processing Unit (CPU). The Design based on Von Neumann Architecture that generally includes Registers, Bus Interface, ALU, Memory and their structures. The CPU Architecture simulation used Logisim which capable to performs the digital logic simulation. Logisim has an ability to perform digital logic to build subcircuit become a larger circuit in a single environment from low-level combinational and sequential circuits to build a complete CPU. An undergraduate student has suffered to study COA with the theory that explained in several lessons, the lesson explained a structure and an architecture computer only. There are not providing a simulation that explained step by step computer architecture. This research to propose design and implementation 8-bit CPU (Central Processing Unit) architecture that designed and implemented step by step that created in Logisim and used the simplest design that is Mic-1. So the simulation easy to visualize then the Undergraduate Student can learn about the next competency which is to design and make a specific purpose computer.

Keywords — 8 bit; CPU Architecture; Logisim; Undergraduate Learning Support

I. INTRODUCTION

Computer Organization and Architecture (COA) is important parts of Computer Engineering [1] and Computer Science [2] body of knowledge that created by IEEE and ACM. The important issues to learning COA is how to define and representing their structure to students [3]. Computer Architecture attribute is computer system associated with programmer such as arithmetic used, set of instruction, addressing technique, the mechanism of the I/O [4]. Then Computer Organization is studying how part of computer associated with the operational unit and their relationship.

Two departments who meet an advance of COA is Computer Science (CS) and Computer Engineering (CE). Based on ACM of CS Curricula and CE Curricula, their body of knowledge is different. COA on CS has 16 core hours [2] and CE has 60 Core hours [1], so that hours will give a huge

impact to what kind and deep of course that given to the student. Based on previous research to earn about details and characteristic of each simulator that provides CPU design that has an ability for learning support on CE student is Logisim [5]. CE have competency to design multipurpose computers so it would be done in CE students [1]. The criteria of selections simulators based on a simulator that can accommodate teaching material and have the capacity to delivers materials to the student. For CE Students includes Computer Arithmetic, Processor Organization, Memory System Organization and Architecture, I/O Interfacing and Communication [6]. Logisim is an educational tool for designing and simulating the digital logic circuit. Even though can only simulate the circuit, Logisim has an ability to simulate each of CPU component. It has a Graphical User Interface (GUI) which students can make, see and modify the digital logic circuit [7].

To teach the student about CPU (Central Processing Unit), it needs a working simulation. The simplest CPU design is 8-bit CPU that was created and called Von Neumann architecture. That design should make sequentially that includes first Simple ALU, Datapath, Control Unit and Integrates all [8]. Several scientists developed CPU with their own approaches, but for students learning purpose needs simple methods that can be understood [9]. The CPU design architecture can make with different approaches. Many commercial computer architecture and organization concept that has been used is 8051 [10], MIPS [11], x86 [12] and ARM architecture [13]. Based on Von Neumann architecture there are developed Mic-1, Mic-2, Mic-3 and Mic-4 architecture. The simplest architecture that used is Mic-1 because the architecture is simple and can accommodate student needs [14].

So in this research, we want to design and implemented a CPU-8 bit design based on Mic-1 architecture. The CPU design should meet student requirement for learning COA on CE. It will be implemented in Logisim which has a GUI, so the student can learn and understand how CPU works, the student can analyze about the detail of how CPU works and student can and make a specific purpose computer. This research focus is how to design CPU and examine its relationship with CE learning outcome.

II. RESEARCH METHOD

A. Course Design based on Learning Outcome

Course design started by specifying Learning Objective, this research use Learning Outcome (LO) that explained CE Computing Curricula [1]. Before giving instruction how LO based on a textbook that used, we need to give an assessment of what tool used. The previous research explained that best tool is Logisim.

To teach the student Computer Organization and Architecture, it needs the simplest design. Other research gives an assessment that the simplest design that could use in learning is used Mic-1 architecture. So this research conducted by studying Learning Objective to give an Assessment only that represent in Fig. 1 [15].

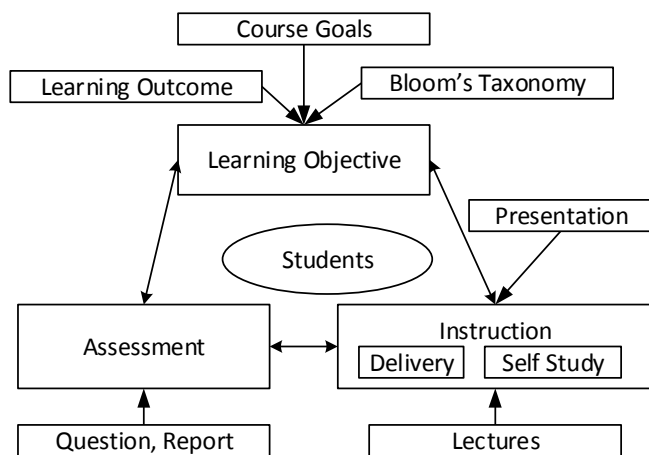


Fig. 1 Learning Method

Learning Objective will be given by study and discussed what kind of LO that should deliver to a student based on Williams Stalling text book. Second, Assessment was made to selecting and designing about 8-bit CPU based on a Mic-1 design that discussed in the previous section. Then after that section, to do is discussed it LO. Computer Engineering – Computer Organization and Architecture Learning Outcome below:

1. History and Overview; has 1 hours to introduce computer development from first until present. Introduce vacuum tube technology, Ultra Large Integration Computer, Integrated Circuit, Embedded Systems with their microprocessor that has been developed.
2. Relevant tools, standards and or engineering constraint; has 1 hours to describe to students how to measure system attributes. How to benchmark system attributes with Amdahl and Little Law.
3. Instruction Set Architecture; has 10 hours this chapter is focused to learn about machine instruction set views by computer designer or programmer. To learn about how that instruction does, to examine the type of

operand and operator. Each of that is specified by machine instruction does.

4. Measuring Performance; has 3 hours. This chapter is extending from the previous chapter that how to systems benchmarking based a common measurement method. The measurement is used to evaluate processor performance based on its architecture.
5. Computer Arithmetic; has 3 hours to learn about Arithmetic and Logic Unit (ALU). To focused leaning ALU arithmetic and their complexity aspect.
6. Processor Organization; has 10 hours. This chapter to introduced and discussed with the student about processor organization from basic until high-level organization. This chapter includes processor organization, register, pipelining until RISC and superscalar architecture.
7. Memory System Organization and Architecture; has 9 hours student learn about internal and external memory. Each of that is about cache memory, ROM, DRAM, SRAM etc. After knowing each memory, a student could discuss how to enhance memory reliability. This chapter includes a magnetic disk, optical memory until the RAID.
8. I/O Interfacing and Communication; has 7 hours to learn I/O module that is the third key of computer element. I/O module represented central switch or controls more than one peripheral device and interfacing to the system bus.
9. Peripheral Subsystem; has 7 hours conducting by peripheral data transfer, the operation method from the processor to control range devices. Different data transfer rate that having by peripheral discussed in this section.
10. Multicore Architecture; has 6 hours that learn about multicore systems. What to do in this chapter first is analyze hardware performing factor, which brings development to the multicore computer system. This chapter includes a multicore product including ARM, Intel, and Mainframes.
11. Distributed System Architecture; has 3 hours this chapter explained an overview of superscalar implementation and their architecture that contrasting by it super-pipelining. This chapter discusses the most architecture that used.

This section would describe the design and implementation 8-bit CPU in Logisim. Andrew Tanenbaum explains Mic-1 architecture [16] about step-by-step designing Mic-1 which has Simple ALU, CPU Datapath, Control unit, and their integration. In order to create CPU, that step cannot be switched or should be sequentially made.

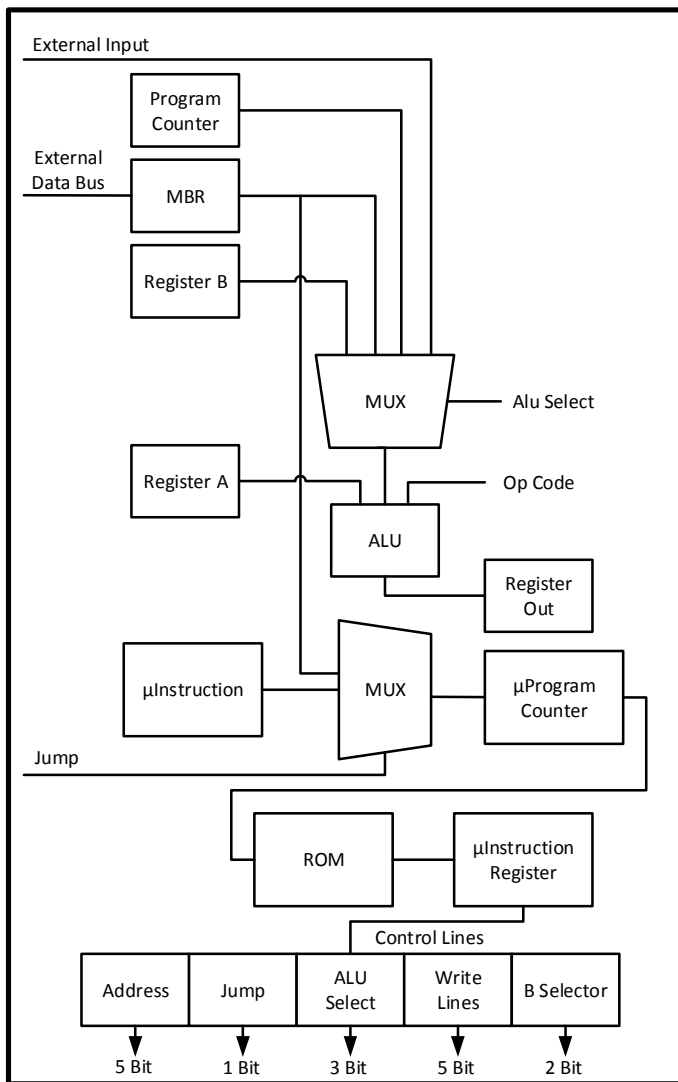


Fig. 2 Mic-1 8-bit CPU Block Diagram

Based on Fig. 2 it would be designed Mic-1 8-bit CPU that doesn't need complex structure. The CPU has registers, program counter, mux, ROM, instruction register etc. It has an external connection which are external input, external data bus, ALU select, and jump. Each component function will be described their detail in this section.

B. A Simple ALU

First, is create ALU that designed based on logic gates. At block diagrams Fig. 3 ALU has two 8-bit bus input that represented by A Bus and B Bus, that ALU is selected by three ALU control Lines that represented by F0, F1, and F2. By three ALU control lines, it can be operated by eight different operations depends on each Boolean combination entered.

ALU can be turned into subcircuit, we just need to create one ALU, and then it can be called if we need ALU function.

A simple ALU that designed only has two registers. First register called Register A used as an accumulator (to save temporary data). The second register called Register B used to process one of four possibilities includes Memory Buffer Register (MBR), Program Counter (PC), Second Accumulator and External Input/output (I/O). This ALU has a number of computation possibilities, as follows: Increment A, Increment B, A+B, A+ (-B), And, Or. ALU cannot be operated by Xor. That computation possibility which selected by MUX, which has a selector for which computation selected.

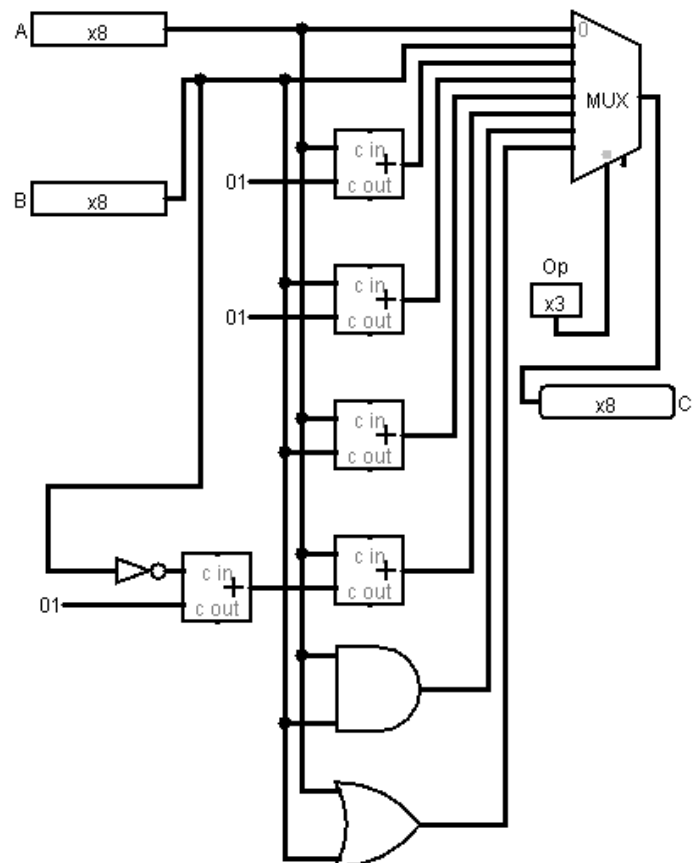


Fig. 3 A Simple ALU Schematic Diagram on Logisim

C. Datapath

The second step is to take ALU inside Datapath. The Datapath design in Fig. 3 based on the representation of basic Von Neumann Architecture. This research used 8-bit bus and registers. The CPU has eventually two general-purpose registers, one output register, Memory Buffer Register (MBR), Program Counter (PC). In this CPU design that functions implemented on Register A.

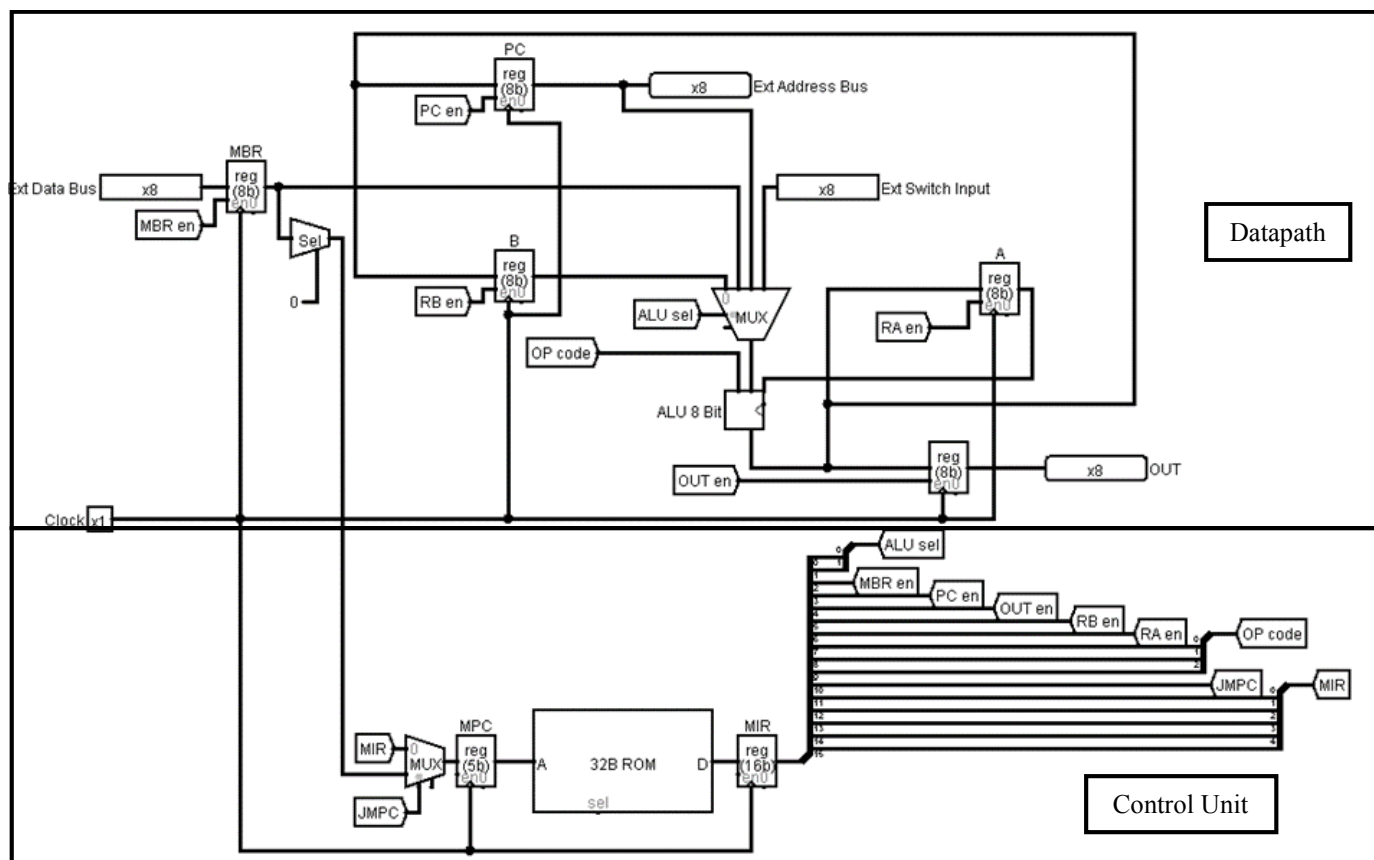


Fig. 4 CPU Datapath and Control Unit Schematic Diagram on Logisim

ALU has two inputs such as register A and MUX. MUX has four possible input, between ALU has two inputs such as register A and MUX. MUX has four possible input, between Memory Buffer Register (MBR), Program Counter (PC), Register B and External Input-Output (I/O). MBR used to save data from Random Access Memory (RAM).

At Logisim, students can simulate CPU Datapath by preloading a register with a preset value. A student could toggles which operation that through ALU by control lines and address lines. The value that presets by the student would appear in Out Register. This process is simulated a “Move” operation in the Datapath. The operation describes the operation sequence of control lines that displayed on the 7-segment display.

D. Control Unit

The control signal in the CPU managed by Datapath which is simulated. The previous step was built ALU and registers to simulate the CPU Datapath. Then, next step is to add the control unit. Machine Language Instruction executed by CPU, so that required a sequence to step through the operation. Various Registers and Control lines in CPU would be managed and coordinated by microprograms so we need ROM that

should be a control store. The control unit is assignments and derived by Mic-1 Architecture which described in the previous section. The control unit based on Mic-1 represented in Fig. 3.

The control unit has a flip-flop, the flip-flop used is a D flip-flop that has two stable states. It can be used to store state information, D stands for "data", D flip-flop is required to pass data. This design uses a flip-flop that has an enabler to activate its active state in the control unit.

Memory Buffer Register (MBR) at Control Unit used to store or put data from each RAM address. After data processed by MBR, the data go to MUX. After that, the data has two chance of process. First chance if the data from MBR include adding, subtract, etc., a second chance is if data from external input. Each has a binary code representing every possible instruction stored by RAM that can give different treatment to the MBR by the Control Unit.

Control Unit task requires 32x16-bit ROM, The 5-bit address bus is connected to Microprograms Program Counter (MPC) that function is by connecting control store. Microprograms Instruction Register (MIR) is special purpose registers that latched by 32-bit or from 16 cables data bus from the control store. This function provides a different enabler state for each D Flip-flop.

Op Code	Bin Addr	Hex Addr	Decimal Addr	Next Addr	addr 1	addr 2	addr 3	addr 4	addr 5	JMPC	ALU 0	ALU 1	ALU 2	RA en	RB en	OUT en	PC en	MBR en	ALU sel 0	ALU sel 1	Comment
(main)	00000000	00	0	1	0	0	0	0	1	0	0	0	1	0	0	0	0	1	1	0	Fetch MBR
	00000001	01	1	2	0	0	0	1	0	0	0	1	1	0	0	0	1	0	1	0	PC = PC + 1
	00000010	02	2	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	Go to MBR
NOP	00000011	03	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Go to (main)
INPUT	00000100	04	4	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	1	Fetch Input to A
OUTPUT	00000101	05	5	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	Move A to Output
JMP <addr>	00000110	06	6	7	0	0	1	1	1	0	0	0	0	0	0	0	0	1	0	0	Fetch MBR

Fig. 5 Sample of Microprograms of Opcode

E. Integration

The integration uses a simple assembly language to test the design. To do this, students must understand the machine language and convert it, so it can be added to the CPU program. The opcode is used, each assembly language used and could be deduced from their starting location/addresses inside the ROM microprograms Fig. 4. Then the student could run several machine language and observe what CPU does. It can be observed with a 7-segment display about how it works. To observe, Logisim have a single step debugging feature and have Clock Speed adjustment feature. So with this feature, a student could observe the behavior of the CPU.

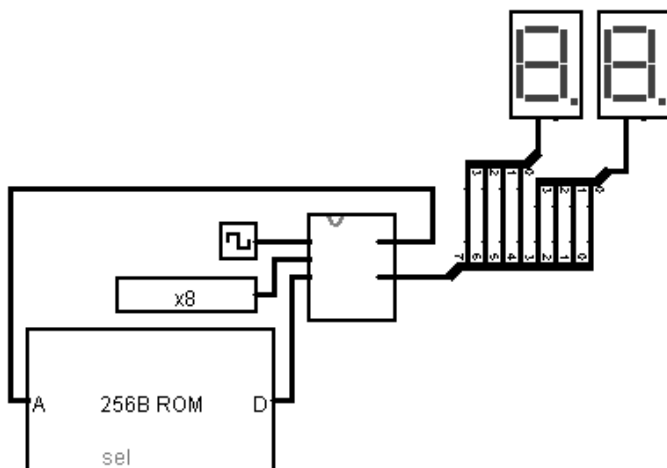


Fig. 6 Final CPU Schematic Diagram at Logisim

Even if assembly language set is limited for instruction, we could add it to a various additional program that assigned. For example to assign an additional program is to create the command that represented with hexadecimal values. Each hexadecimal values could switches and display their values into the 7-segment display, or integrated with input/output device and program memory at Fig. 6.

III. RESULTS AND DISCUSSION

In this section will be given the results of a discussion that connects the LO on the CE of the COA and its relationship with 8-bit CPU. So that will be obtained suitability between design 8-bit CPU with LO at CE. This section is to discuss how to reach Learning Outcome at Computer Engineering based on 8-bit CPU design at Logisim.

1. Understand the difference between computer organization and architecture computer. On Logisim Computer Organization represent the instruction set to represent data type, for example, hexadecimal or binary data. Computer Architecture to represent data used to represent their operational units and their interconnection, that data representation called opcode.
2. Visualize computer main component. On Logisim performed ALU, Datapath and Control Unit which inside ALU and Control Unit could see their schematic diagram. On Datapath could see by their block diagrams.
3. Computer architecture has their strength and weaknesses based on reasons and strategies. On Logisim performed by the numbers of bits that saved by the opcode. The opcode has the relationship with the amount of syntax which configured. Syntax represents with assembly language that could perform by computer. The amount of bit data represent how much syntax that could run by computer.
4. Identify high-performance computing by any techniques such as parallel and distributed computer architecture. On Logisim Parallel processing could represent by ALU, it will call Parallel if the computer has two or more ALU that runs the different process at same time. The distributed computer could represent by two or more Control unit which runs the same process at same time.
5. Computer architecture and organization has benefits to computer engineering. So, at Logisim student can build a specific multipurpose computer that made according to by the specific requirement.

IV. FUTURE WORK

LO on CE student is very deep than CS, so to complete the lesson is very different. Therefore this paper is to present how COA on CE is done by designing and implementing 8-bit CPU on Logisim. The student learns about how CPU work on a different dimension. To extend in the future Course design based on learning method which involved by the lecturer or by the student to make an assessment about how suitable Logisim to perform learning support. If we get a good responder, then the Logisim will be recommended as a learning support tool or could be a tool to design or enhance CPU until designing specific purpose computer.

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